



UNIVERSITÀ
DI TRENTO



Master's Degree in Cognitive Science

NEURAL MASS AND SPIKING MODELS FOR THE STUDY
OF NEURAL CIRCUITS

Supervisor:
Albrecht Haase

Student:
Zeno Rossi

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Contents

- 1 Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe** **1**
 - 1.1 Introduction 1
 - 1.1.1 Classical Stochastic Resonance 1
 - 1.1.2 From Earth to neurons 4
- A Stability of Equilibria** **5**

Abstract

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Chapter 1

Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1–3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical Stochastic Resonance

The idea of studying the effects of noise on the dynamics of a system first appeared parallely in the work of Benzi et al. [5] and Nicolis [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature.

A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C \frac{dT}{dt} = R_{in}(T) - R_{out}(T) \quad (1.1)$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$R_{in} = Q\mu \quad (1.2)$$

$$R_{out} = \alpha(T)Q\mu + \varepsilon(T) \quad (1.3)$$

where μ is the average incoming solar radiation.

In [5] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

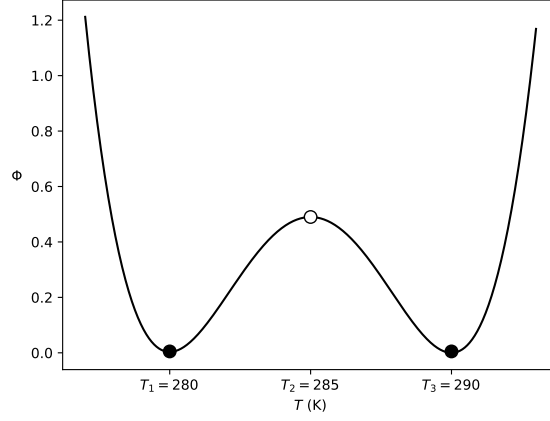


Figure 1.1: The double well potential. Stable points filled

$$C \frac{dT}{dt} = \frac{\mu \varepsilon(T)}{1 + \beta(1 - T/T_1)(1 - T/T_2)(1 - T/T_3)} - \varepsilon(T) \quad (1.4)$$

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{dT}{dt}, \quad (1.5)$$

then a pseudo-potential energy function is defined as

$$\Phi(T) = - \int F(T) dT \quad (1.6)$$

that represents the "energy" needed to change the temperature across temperature values. From Φ we get

$$F(T) = - \frac{\partial \Phi}{\partial T} \quad (1.7)$$

Then intuitively the maximum of Φ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{dF}{dT} = - \frac{\partial^2 \Phi}{\partial T^2} \quad (1.8)$$

is < 0 , any neighboring starting state of the earth's temperature will converge to it, while when it is > 0 they will diverge.

It can be shown that T_1, T_3 are stable points, while T_2 is unstable: a peak any starting state would "roll off" from (see fig). In our context, T_1, T_3 are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. This type of system where 2 stable states separated by an unstable state exist is called a bistable system, and the plot of Φ over T displays the typical double well shape of the (pseudo)potential function (see Figure 1.1).

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., whether it is smaller or greater than T_2 . Then it would follow $F(T)$ towards T_1 or T_3 (see Figure 1.2). However, earth's temperatures do indeed show oscillations with a period of 10^5 years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then μ becomes time dependent

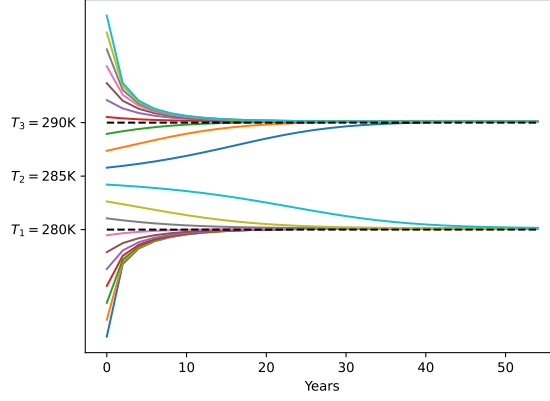


Figure 1.2: Relaxation for different starting temperatures without noise

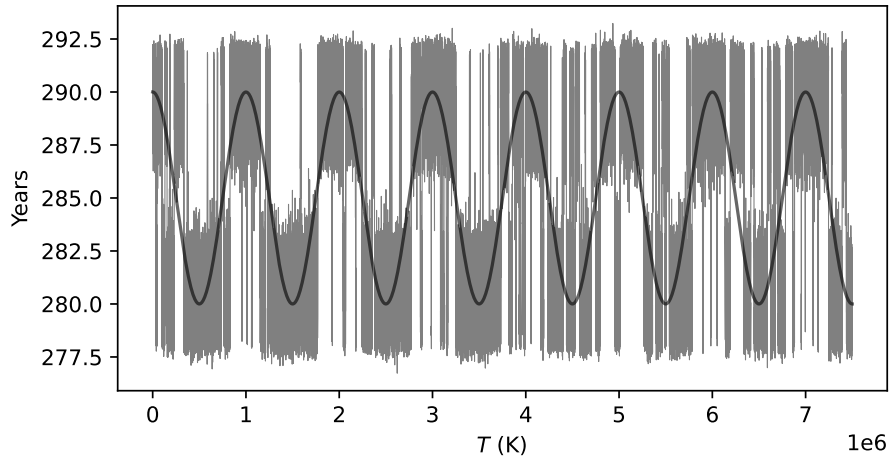


Figure 1.3: Temperature over time, with noise

$$\mu(t) = 1 + \gamma \cos(\omega t) \quad (1.9)$$

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become time dependent $\tilde{F} = \tilde{F}(T; t)$, $\tilde{\Phi} = \tilde{\Phi}(T; t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over T_2 (see fig).

If gaussian white noise, representing various mechanisms causing T fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{dT}{dt} = F(T) + \sigma \eta(t) \quad (1.10)$$

then for certain values of σ a jump between stable states becomes possible, with a frequency $\approx 2\pi/10^5$. That is, the system shows temperature oscillations at the forcing frequency, as noise allows the system to jump from one equilibrium to another (Figure 1.3). Note that oscillations arise from the joint effect of the periodic forcing and noise: noise pushes the system away from the equilibrium at every timestep, but only when the forcing asymmetrically tilts the double well potential such that the distance between the current equilibrium and the unstable one becomes small enough to allow the stochastic component to push the system over it, a climate transition (that is, a jump from T_1 to T_3 or viceversa) can occur. This can be seen as an "escape" from a stable equilibrium, requiring a jump over a (pseudo)energy potential "wall". This is a

classic problem in physics, Kramer's rate [7] gives the frequency of jumps per second

$$r_K = \frac{\sqrt{V''(x_m)/m} \sqrt{|V''(x_b)/m|}}{2\pi\gamma} e^{-\frac{\Delta V}{D}} \quad (1.11)$$

where $V''(x_m)$ is the curvature around the minima, $V''(x_b)$ that at the maximum, ΔV the distance from the minimum to the maximum, and $D = \sigma^2/2$ the temperature of the system (noise intensity in our context). The inverse $\frac{1}{r_K}$ gives the residence time, the time spent at one of the minima. If our condition for SR is that it allows the system to follow the subthreshold forcing, then the optimal D is such that, given the forcing period T_ω and the residence time as a function of D , $T_K(D) = \frac{1}{r_K(D)}$ is

$$T_K(D) = \frac{T_\omega}{2} \quad (1.12)$$

Some useful considerations can be made on SR defined on the climate model. First, it requires a system containing a nonlinearity, as in 1.4, such that 2 stable states separated by a potential wall exist. Additionally, the periodic forcing must be insufficient alone to cause a crossing (subthreshold), though later it will be shown that the forcing assumptions can be relaxed. Last, as 1.12 suggests, too low D will not be able to push the system over the barrier at the frequency needed, and high enough D make the potential barrier effectively nonexistent, increasing r_K and making the double well model meaningless.

Hence, given a metric of how well the system follows the forcing, there will be a peak for a certain D . In this sense, there is a resonance in the system. For classic SR models, the natural measure is the spectral power (obtained by Fourier transform of the system response) at the forcing frequency relative to the neighboring ones: Signal-to-Noise Ratio (SNR) [8].

1.1.2 From Earth to neurons

Appendix A

Stability of Equilibria

In Benzi et al. [5]

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